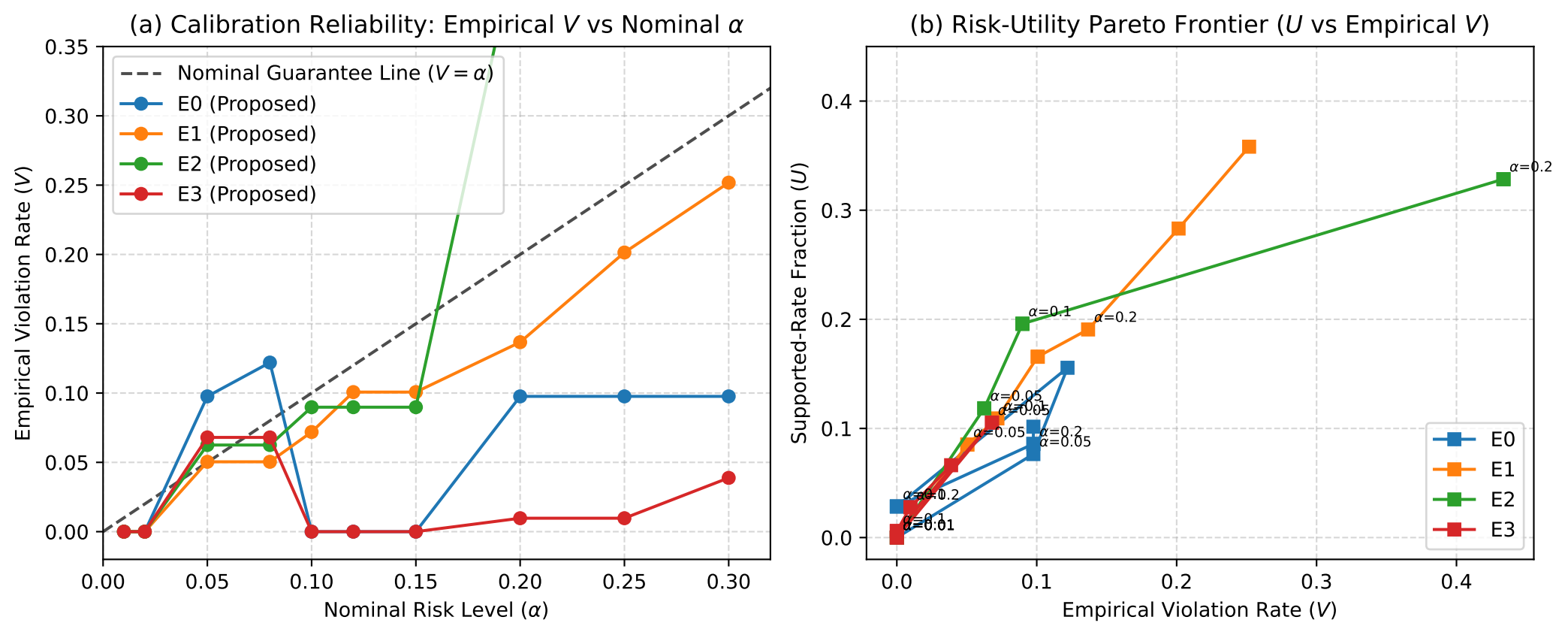
(a) Calibration Reliability: Empirical V vs Nominal α
Nominal Guarantee Line ($V = \alpha$)
E0 (Proposed)
E1 (Proposed)
E2 (Proposed)
E3 (Proposed)
Empirical Violation Rate (V)
Nominal Risk Level (α)
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
(b) Risk-Utility Pareto Frontier (U vs Empirical V)
Supported-Rate Fraction (U)
Empirical Violation Rate (V)
0.0
0.1
0.2
0.3
0.4
E0
E1
E2
E3
α=0.2
α=0.1
α=0.2
α=0.05
α=0.05
α=0.05
α=0.2
α=0.05
α=0.2
α=0.1
α=0.01